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IT342-G3
SYSTEMS INTEGRATION AND ARCHITECTURE 1

SOFTWARE TEST PLAN (STP)

Project Title: WearIsIt

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Introduction

WearIsIt is a web and mobile application that allows users to manage clothing items and outfits digitally. Phase 1 focuses on delivering core functionality for user access and basic wardrobe management.

The test team is responsible for validating the system and also acts as the **end user**, ensuring the application meets real-world usability needs.

Functional Requirements Coverage

Authentication (Web & Mobile)

- Register with valid/invalid inputs
- Login with correct/incorrect credentials
- Logout functionality

Google OAuth (Web)

- Login/Register using Google account
- Handle denied permissions
- Verify account linking

Clothing Management (Web)

- Create clothing item with valid/invalid data
- Upload image (supported formats, size limits)
- View gallery (list loads correctly)
- View item details (correct data displayed)

Test Execution

1. Test Execution Approach

Test execution will be carried out after test cases have been designed and the test environment is stable. The execution process will involve running both automated backend tests and manual tests for the web and mobile applications.

1.1 Initial System Load Verification

Each system build will be initially launched to verify that the application starts correctly and that no critical startup errors are present. This ensures the environment is stable before formal test execution begins.

1.2 Backend Testing (Unit Testing)

Automated unit tests will be executed for backend modules. These tests validate individual components in isolation to ensure correct logic, data handling, and system behavior.

1.3 Web Application Testing (Manual Execution)

The test team will manually execute all defined web test cases. This includes:

- User authentication (including Google OAuth)
- Clothing item creation and upload
- Gallery and item detail verification

Actual results will be compared against expected results and recorded accordingly.

1.4 Mobile Application Testing (Manual Execution)

Manual test execution will be performed on the mobile application. The focus will be on:

- User registration
- User login
- User logout

Each test case will be executed step-by-step and results documented.

1.5 Defect Logging

Any deviation between expected and actual results will be logged as a defect. Defects will be categorized based on severity and tracked until resolution and retesting.

Entry Criteria

Test execution will begin only when the following conditions are met:

- All required features for the test phase are implemented
- Test cases are fully prepared and reviewed
- Test environment (web and mobile) is stable and accessible
- Backend services and database are running correctly

- Test data is available and prepared
- No critical blocking issues remain open from development

Exit Criteria

Test execution will be considered complete when:

- All planned test cases have been executed
- All critical and high-severity defects are resolved and retested
- Remaining defects are documented and accepted (if any)
- Core functionalities (login, registration, OAuth, upload, gallery) are stable
- Test results have been recorded and reviewed
- A Full Regression Test Report has been produced

Test Cases

TS-001: Authentication Module

Test Case ID	Platform	Test Scenario	Preconditions	Test Steps	Expected Result
TC-AUTH-001	Web	Load registration page	App is accessible	Open registration page	Registration form loads successfully
TC-AUTH-002	Mobile	Load registration screen	App installed	Open registration screen	Registration form displays correctly
TC-AUTH=003	Web/Mobile	Submit empty registration form	On registration page	Click submit without input	Validation errors shown
TC-AUTH-004	Web/Mobile	Invalid email format	On registration page	Enter invalid email → submit	Email validation error displayed

TC-AUTH-005	Web/Mobile	Successful registration	Valid user data	Fill form → submit	User created and redirected to gallery/profile page
TC-AUTH-006	Web/Mobile	Successful login	Registered user exists	Enter valid credentials → login	User logged in successfully
TC-AUTH-007	Web/Mobile	Invalid login	Registered user exists	Enter wrong password → login	Error message displayed
TC-AUTH-008	Web/Mobile	Logout	User logged in	Click logout	User redirected to login and token is cleared on client side

TS-002: Clothing Item Upload Module

Test Case ID	Platform	Test Scenario	Preconditions	Test Steps	Expected Result
TC-ITEM-001	Web	Access upload page	User logged in	Navigate to upload page	Upload form loads correctly
TC-ITEM-002	Web	Select clothing category	On upload form	Choose category/type	Selected category is reflected
TC-ITEM-003	Web	Upload valid image	On upload form	Upload .jpg/.png file	Image preview displayed
TC-ITEM-004	Web	Upload invalid file type	On upload form	Upload unsupported file	Error message shown
TC-ITEM-005	Web	Submit valid clothing item	All fields filled	Submit form	Item saved successfully

TC-ITEM-006	Web	Submit empty form	On upload form	Submit without data	Validation errors displayed
TC-ITEM-007	Web	Verify item appears in gallery	Item created	Go to gallery	New item visible in gallery

TS-003: Gallery Module

Test Case ID	Platform	Test Scenario	Preconditions	Test Steps	Expected Result
TC-GRID-001	Web	Load gallery page	User logged in	Navigate to gallery	Gallery loads successfully
TC-GRID-002	Web	Display user-specific items	Items exist	Open gallery	Only user's items are shown
TC-GRID-003	Web	Verify item details from gallery	Items exist	Click item	Correct item details displayed
TC-GRID-004	Web	Empty gallery state	No items exist	Open gallery	Empty state message shown

TS-003: OAuth Log In Module

Test Case ID	Platform	Test Scenario	Preconditions	Test Steps	Expected Result
TC-OAUTH-001	Web	Display Google Login option	User is on login page	Open login page	"Login with Google" button is visible

TC-OAUTH-002	Web	Successful Google Login	Valid Google account	Click “Login with Google” → Select account → Grant permissions	User is logged in and redirected to gallery
TC-OAUTH-003	Web	Cancel Google Login	On Google login screen	Click “Login with Google” → Cancel authentication	User returned to login page, no login occurs
TC-OAUTH-004	Web	Logout after Google Login	User logged in via Google	Click logout	User logged out and redirected to login page
TC-OAUTH-005	Web	Re-login with same Google account	User already registered via Google	Logout → Login again via Google	Same account is used, no duplicate created

Test Scripts / Test Steps

TC-AUTH-001: Load Registration Page (Web)

Field	Details
Test Script ID	TS-AUTH-001
Platform	Web
Test Scenario	Load registration page
Preconditions	Application is accessible
Test Steps	1. Open browser 2. Navigate to WearIsIt URL 3. Click “Register” button

Expected Result	Registration form loads successfully with all required fields visible

TC-AUTH-002: Load Registration Screen (Mobile)

Field	Details
Test Script ID	TS-AUTH-002
Platform	Mobile
Test Scenario	Load registration screen
Preconditions	App installed on device
Test Steps	1. Open WearIsIt mobile app 2. Click register link
Expected Result	Registration screen displays correctly

TC-AUTH-003: Submit Empty Registration Form

Field	Details
Test Script ID	TS-AUTH-003
Platform	Web/Mobile
Test Scenario	Submit empty registration form
Preconditions	User is on registration page
Test Steps	1. Open registration form 2. Leave all fields blank 3. Click “Submit”
Expected Result	Validation errors displayed for required fields

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TC-AUTH-004: Invalid Email Format

Field	Details
Test Script ID	TS-AUTH-004
Platform	Web/Mobile
Test Scenario	Validate email format
Preconditions	User is on registration page
Test Steps	1. Enter invalid email (e.g. `user@`) 2. Fill other required fields 3. Click “Submit”
Expected Result	Email format validation error is displayed

TC-AUTH-005: Successful Registration

Field	Details
Test Script ID	TS-AUTH-005
Platform	Web/Mobile
Test Scenario	Register new user successfully
Preconditions	Valid user data available
Test Steps	1. Open registration form 2. Enter valid name, email, password 3. Click “Submit”
Expected Result	User account is created and redirected to gallery/profile page

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TC-AUTH-006: Successful Login

Field	Details
Test Script ID	TS-AUTH-006
Platform	Web/Mobile

Test Scenario	User login with valid credentials
Preconditions	Registered user exists
Test Steps	1. Open login page 2. Enter valid email and password 3. Click “Login”
Expected Result	User is authenticated and redirected to home/gallery page

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TC-AUTH-007: Invalid Login

Field	Details
Test Script ID	TS-AUTH-007
Platform	Web/Mobile
Test Scenario	Login with incorrect credentials
Preconditions	Registered user exists
Test Steps	1. Open login page 2. Enter valid email + wrong password 3. Click “Login”
Expected Result	Error message displayed (invalid credentials)

TC-AUTH-008: Logout

Field	Details
Test Script ID	TS-AUTH-008
Platform	Web/Mobile
Test Scenario	User logout
Preconditions	User is logged in
Test Steps	1. Click “Logout” button 2. Observe session state

Expected Result	User is redirected to login page and authentication token/session is cleared

TC-ITEM-001: Access Upload Page

Field	Details
Test Script ID	TS-ITEM-001
Platform	Web
Test Scenario	Access upload page
Preconditions	User is logged in
Test Steps	1. Open browser 2. Log in to WearIsIt 3. Navigate to upload page
Expected Result	Upload form loads correctly

TC-ITEM-002: Select Clothing Category

Field	Details
Test Script ID	TS-ITEM-002
Platform	Web
Test Scenario	Select clothing category
Preconditions	Upload page is open
Test Steps	1. Open category dropdown 2. Select clothing type (e.g. Shirt, Pants)
Expected Result	Selected category is reflected in UI

TC-ITEM-003: Upload Valid Image

Field	Details
Test Script ID	TS-ITEM-003
Platform	Web
Test Scenario	Upload valid image
Preconditions	Upload form is open
Test Steps	1. Click upload button 2. Select valid .jpg or .png file 3. Confirm upload
Expected Result	Image preview is displayed before submission

TC-ITEM-004: Upload Invalid File Type

Field	Details
Test Script ID	TS-ITEM-004
Platform	Web
Test Scenario	Upload unsupported file type
Preconditions	Upload form is open
Test Steps	1. Click upload 2. Select invalid file (e.g. .exe or .gif if not allowed) 3. Attempt upload
Expected Result	Error message is displayed

TC-ITEM-005: Submit Valid Clothing Item

Field	Details
Test Script ID	TS-ITEM-005
Platform	Web

Test Scenario	Submit valid clothing item
Preconditions	All fields filled correctly
Test Steps	1. Fill all required fields 2. Upload valid image 3. Click submit
Expected Result	Item is saved successfully

TC-ITEM-006: Submit Empty Form

Field	Details
Test Script ID	TS-ITEM-006
Platform	Web
Test Scenario	Submit empty upload form
Preconditions	On upload page
Test Steps	1. Leave all fields blank 2. Click submit
Expected Result	Validation errors are displayed

TC-ITEM-007: Verify Item in Gallery

Field	Details
Test Script ID	TS-ITEM-007
Platform	Web
Test Scenario	Verify uploaded item appears in gallery
Preconditions	Item has been created
Test Steps	1. Navigate to gallery page 2. Check item list
Expected Result	Newly uploaded item is visible in gallery

TC-GRID-001: Load Gallery Page

Field	Details
Test Script ID	TS-GRID-001
Platform	Web
Test Scenario	Load gallery page
Preconditions	User is logged in
Test Steps	1. Open WearIsIt web app 2. Log in with valid credentials 3. Navigate to gallery page
Expected Result	Gallery page loads successfully with UI displayed

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TC-GRID-002: Display User-Specific Items

Field	Details
Test Script ID	TS-GRID-002
Platform	Web
Test Scenario	Display only user-owned clothing items
Preconditions	User has existing uploaded items
Test Steps	1. Log in to account 2. Navigate to gallery page 3. View displayed items
Expected Result	Only items belonging to the logged-in user are shown

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TC-GRID-003: View Item Details

Field	Details
Test Script ID	TS-GRID-003

Platform	Web
Test Scenario	Verify clothing item details from gallery
Preconditions	At least one item exists in gallery
Test Steps	1. Open gallery page 2. Click on a clothing item 3. View item details page
Expected Result	Correct item details (name, type, image) are displayed

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TC-GRID-004: Empty Gallery State

Field	Details
Test Script ID	TS-GRID-004
Platform	Web
Test Scenario	Handle empty gallery state
Preconditions	User has no uploaded items
Test Steps	1. Log in with user account 2. Navigate to gallery page
Expected Result	Empty state message is displayed (e.g., “No items found”)

TC-OAUTH-001: Display Google Login Option

Field	Details
Test Script ID	TS-OAUTH-001
Platform	Web
Test Scenario	Display Google Login button
Preconditions	User is on login page
Test Steps	1. Open WearIsIt login page

Expected Result	“Login with Google” button is visible

TC-OAUTH-002: Successful Google Login

Field	Details
Test Script ID	TS-OAUTH-002
Platform	Web
Test Scenario	Successful Google authentication login
Preconditions	Valid Google account exists
Test Steps	1. Click “Login with Google” 2. Select Google account 3. Grant permissions
Expected Result	User is authenticated and redirected to gallery page with tokens issued

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TC-OAUTH-003: Cancel Google Login

Field	Details
Test Script ID	TS-OAUTH-003
Platform	Web
Test Scenario	Cancel OAuth authentication flow
Preconditions	User initiates Google login
Test Steps	1. Click “Login with Google” 2. Cancel Google authentication
Expected Result	User is redirected back to login page and no session is created

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TC-OAUTH-004: Logout After Google Login

Field	Details
Test Script ID	TS-OAUTH-004
Platform	Web
Test Scenario	Logout after OAuth login
Preconditions	User is logged in via Google OAuth
Test Steps	1. Click logout button
Expected Result	User session is cleared and redirected to login page

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TC-OAUTH-005: Re-login with Same Google Account

Field	Details
Test Script ID	TS-OAUTH-005
Platform	Web
Test Scenario	Re-authenticate using same Google account
Preconditions	User already exists in system via Google login
Test Steps	1. Logout 2. Click “Login with Google” 3. Sign in with same account
Expected Result	Same user account is reused, no duplicate user is created

Automated Test Cases

Log In Unit Tests

Test Case ID	Test Method	Test Scenario	Test Description	Expected Result
TC-AUTH-AUT-001	`login_ValidUsername_ShouldReturnTokens`	Login using valid username	User logs in using correct username and password	Returns access token, refresh token, and user details
TC-AUTH-AUT-002	`login_ValidEmail_ShouldReturnTokens`	Login using valid email	User logs in using email instead of username	Returns valid authentication tokens and user data
TC-AUTH-AUT-003	`login_UserNotFound_ShouldThrowAuth001`	Login with non-existent user	System attempts login with unknown username/email	Throws `AUTH-001` (Invalid credentials)
TC-AUTH-AUT-004	`login_WrongPassword_ShouldThrowAuth001`	Login with incorrect password	User exists but password does not match	Throws `AUTH-001` (Invalid credentials)
TC-AUTH-AUT-005	`login_EmailFallbackLogic_ShouldUseEmailSearch`	Email fallback authentication	System uses email search when username lookup fails	Successfully authenticates using email and returns tokens

Register Unit Tests

Test Case ID	Test Method	Test Scenario	Test Description	Expected Result
TC-AUTH-AUT-011	`register_ValidFields_ShouldSucceed`	Successful user registration	User registers with valid email, username, and password	User is created and access/refresh tokens are returned
TC-AUTH-AUT-012	`register_EmptyEmail_ShouldThrowAuth006`	Empty email validation	User submits registration with empty email field	Throws `AUTH-006` (Invalid email format)
TC-AUTH-AUT-013	`register_NullEmail_ShouldThrowAuth006`	Null email validation	User submits registration with null email	Throws `AUTH-006` (Invalid email format)
TC-AUTH-AUT-014	`register_InvalidEmailFormat_ShouldThrowAuth006`	Invalid email format	User submits incorrectly formatted email	Throws `AUTH-006`
TC-AUTH-AUT-015	`register_EmptyPassword_ShouldThrowAuth007`	Empty password validation	User submits empty password	Throws `AUTH-007` (Password cannot be empty)
TC-AUTH-AUT-016	`register_NullPassword_ShouldThrowAuth007`	Null password validation	User submits null password	Throws `AUTH-007`

TC-AUTH-AUT-017	`register_EmptyUsername_ShouldThrowAuth008`	Empty username validation	User submits empty username	Throws `AUTH-008` (Username cannot be empty)
TC-AUTH-AUT-018	`register_NullUsername_ShouldThrowAuth008`	Null username validation	User submits null username	Throws `AUTH-008`
TC-AUTH-AUT-019	`register_EmailAlreadyExists_ShouldThrowAuth004`	Duplicate email check	User registers with existing email	Throws `AUTH-004` (Email already registered)
TC-AUTH-AUT-020	`register_UsernameAlreadyExists_ShouldThrowAuth009`	Duplicate username check	User registers with existing username	Throws `AUTH-009` (Username already exists)

Log Out Unit Test

Test Case ID	Test Method	Test Scenario	Test Description	Expected Result
TC-AUTH-AUT-021	`logout_ShouldRevokeRefreshToken`	User logout process	User logs out using a valid refresh token	Refresh token is revoked successfully and session is invalidated

Clothing Item Unit Tests

Test Case ID	Test Method	Test Scenario	Test Description	Expected Result
TC-ITEM-AUT-001	createClothingItem_Success_ShouldSaveItem	Successful item creation	User creates a clothing item with valid type	Item is created and response contains correct item name and type
TC-ITEM-AUT-002	createClothingItem_InvalidType_ShouldThrowITEM003	Invalid item type	User provides non-existent type ID	Throws `ITEM-003 (Item type not found)` and item is not saved
TC-ITEM-AUT-003	createClothingItem_WithCategories_ShouldAttachCategories	Category attachment	User creates item with category IDs	Item is created and categories are linked correctly
TC-ITEM-AUT-004	createClothingItem_WithTags_ShouldAttachTags	Tag attachment	User creates item with tag IDs	Item is created and tags are linked correctly
TC-ITEM-AUT-005	getClothingItem_Success_ShouldReturnItem	Retrieve existing item	User requests valid item ID they own	Returns correct clothing item details
TC-ITEM-AUT-006	getClothingItem_NotFound_ShouldThrowITEM001	Item not found	User requests invalid or unauthorized item ID	Throws `ITEM-001 (Item not found)`

TC-ITEM-AUT-007	getUserClothingItems_ShouldReturnList	Retrieve user items	User fetches all their clothing items	Returns list of items (non-empty when data exists)
TC-ITEM-AUT-008	getUserClothingItems_EmptyList_ShouldReturnEmpty	Empty dataset handling	User has no clothing items	Returns empty list without error

Image Upload Unit Tests

Test Case ID	Test Method	Test Scenario	Test Description	Expected Result
TC-IMG-AUT-001	uploadImage_validFile_shouldSucceed	Successful image upload	Valid image file is uploaded to storage service	Returns a non-null URL containing correct Supabase storage path and calls RestTemplate once
TC-IMG-AUT-002	uploadImage_fileTooLarge_shouldThrowFILE001	File size validation	User uploads file larger than 10MB	Throws `FILE-001 (File size exceeds limit)` and no upload request is sent
TC-IMG-AUT-003	uploadImage_invalidFileType_shouldThrowFILE002	File type validation	User uploads unsupported file type (e.g., .txt)	Throws `FILE-002 (Invalid file type)` and no upload request is sent